📋 TEST STRATEGY DOCUMENT

Salesforce CPQ Enhancement – Wealth Management Financial Project

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Prepared By: Senior Test Architect

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1. EXECUTIVE SUMMARY

This Test Strategy defines the comprehensive testing approach for the enhancement of an existing Salesforce CPQ implementation supporting Wealth Management services. The strategy adopts a risk-based, shift-left testing methodology with emphasis on quote calculation accuracy, integration data integrity, and regulatory compliance (GDPR, audit trails). The document outlines scope, test types, environments, automation strategy, data management, risk mitigation, and exit criteria to ensure a controlled, compliant, and high-quality release.

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2. PROJECT CONTEXT & SCOPE

2.1 Project Overview

Attribute	Details
Project Type	Enhancement of Existing Salesforce CPQ
Business Domain	Wealth Management Financial Services
In-Scope CPQ Modules	Product Configuration, Pricing Rules, Quote Generation, Contract Management
Out of Scope	None – Full end-to-end validation required

2.2 Key Business Objectives

- Enable accurate, compliant quote generation for complex wealth management products

- Ensure seamless integration with SAP (financial posting), KYC (compliance), DocuSign (contract execution), and CRM Analytics

- Maintain GDPR compliance with robust audit trails for all quote/contract modifications

- Support quarter-end peak loads with <10 minute quote generation SLA

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3. TEST SCOPE & OBJECTIVES

3.1 In-Scope Testing Areas

✅ Functional Testing

- Product configuration rules (bundled products, tiered pricing, eligibility)
- Pricing engine validation (discounts, commissions, multi-currency calculations)
- Quote lifecycle: Draft → Approval → Generation → Contract → Renewal/Amendment
- RBAC validation across 8-10 wealth management roles (Advisor, Compliance, Finance, Ops, Admin, etc.)

✅ Integration Testing

- CPQ ↔ Salesforce Sales Cloud (lead/opportunity sync)
- CPQ ↔ SAP (financial posting, GL reconciliation)
- CPQ ↔ KYC/AML systems (real-time compliance validation)
- CPQ ↔ DocuSign (envelope creation, status sync, signed document retrieval)
- CPQ ↔ CRM Analytics (quote performance dashboards)

✅ Non-Functional Testing

- Performance: Quote generation <10 min SLA under quarter-end concurrent load
- Security: GDPR data masking, field-level security, RBAC isolation
- Reliability: Audit trail completeness for all quote/contract state changes
- Data Integrity: Reconciliation validation between CPQ quotes and SAP postings

✅ Compliance & Regulatory Testing

- GDPR: PII masking in test data, right-to-erasure validation
- Audit Logging: Immutable trail of who changed what, when, and why
- Financial Controls: Segregation of duties, approval workflow enforcement

3.2 Out-of-Scope (Explicitly Excluded)

✕ WCAG/Accessibility testing (per business decision)

✕ Mobile responsiveness validation

✕ Load testing beyond quarter-end peak scenarios

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4. TEST APPROACH & METHODOLOGY

4.1 Testing Pyramid Strategy

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┌──────────────────┐
│   E2E/UAT      │ ← 10% - Business workflow validation
├──────────────────┤
│ Integration    │ ← 30% - API contract & data flow tests
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│  Component     │ ← 60% - Provar automation: CPQ rules, UI, APIs
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4.2 Test Types & Coverage Matrix

Test Type	Tool/Framework	Coverage Target	Automation %	Trigger
Unit/Component	Provar, Apex Tests	CPQ rules, pricing logic	80%	Developer Commit
API Integration	Provar, Postman + Newman	REST/SOAP contracts	70%	CI Pipeline
System/E2E	Provar + Synthetic Data	Quote-to-Cash workflows	60% (target)	SIT Deployment
Performance	JMeter + Salesforce Inspector	Quote SLA, quarter-end load	100%	Pre-UAT Gate
Security/RBAC	Manual + Provar	Role isolation, field security	40%	SIT/UAT
Compliance/Audit	Manual Validation + Logs	GDPR, audit trail completeness	100%	UAT Sign-off
Reconciliation	SQL/SAP Queries + Scripts	CPQ ↔ SAP data consistency	50%	Post-Deployment

4.3 Risk-Based Testing Prioritization

Priority 1 (Critical Path):

- Quote calculation accuracy for complex bundled products with tiered pricing

- Multi-currency/multi-entity quote scenarios with FX rate validation

- Integration data integrity: CPQ → SAP financial posting reconciliation

Priority 2 (High Impact):

- RBAC validation: Ensure Advisors cannot view/approve quotes outside their hierarchy

- Audit trail completeness: Every field change logged with user/timestamp/reason

- DocuSign sync failure handling: Graceful degradation and retry mechanisms

Priority 3 (Compliance/Operational):

- GDPR masking validation across all environments

- Partial approval workflows with compliance hold states

- CRM Analytics data accuracy from CPQ source

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5. TEST ENVIRONMENTS & DATA STRATEGY

5.1 Environment Landscape

Environment	Purpose	Data Strategy	Refresh Frequency
Dev	Unit/Component testing	Synthetic data + masked subsets	On-demand
SIT	Integration/E2E testing	Masked production copy + synthetic	Weekly
UAT	Business validation	GDPR-compliant masked production	Per UAT cycle
Pre-Prod	Performance baseline & smoke	Anonymized production-like volume	Pre-release

5.2 Test Data Management (TDM)

- Masking Approach: Salesforce native anonymization for PII fields (Name, Email, SSN, Account Numbers)

- Synthetic Data Generation: Used for:

  - Edge cases: Extreme discount tiers, multi-currency combinations

  - Negative testing: Invalid KYC responses, DocuSign timeout scenarios

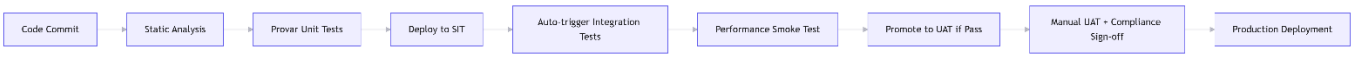
- Data Subset Strategy: Extract relevant wealth management product families + customer segments to reduce environment footprint while maintaining test coverage

6. AUTOMATION & CI/CD INTEGRATION

6.1 Automation Framework

- Primary Tool: Provar (Salesforce-native test automation)
- Scope Expansion: Increase automation coverage from 30% → 60% for regression suite
- API Testing: REST/SOAP integration tests automated via Provar + Postman collections
- Reusable Components:
  - Quote generation test harness
  - SAP reconciliation validator
  - DocuSign status sync monitor

6.2 CI/CD Pipeline (Jira-Driven)



Quality Gates:

- ❌ Block promotion if: Critical/High defects open OR automation pass rate <95%
- ⚠️ Warn if: Performance SLA breached OR audit trail gaps detected

7. DEFECT MANAGEMENT & EXIT CRITERIA

7.1 Defect Severity Classification

Severity	Definition	UAT Exit Requirement
Critical	System crash, data loss, compliance	ZERO tolerated
High	Major function broken, workaround	ZERO tolerated
Medium	Minor function impaired, no financia	<5 open, with mitigation plan
Low	Cosmetic/UI issues, documentation g	No block on sign-off

7.2 Formal Sign-off Requirements

- Primary Stakeholder: Business Product Owner (Wealth Management)
- Mandatory Approvers:
  - Compliance Officer (GDPR/audit validation)
  - Finance Representative (SAP reconciliation sign-off)

- Evidence Required:
  - Test execution report with 100% Priority 1 test coverage
  - Performance baseline report confirming <10 min quote SLA
  - Audit trail sample validation report
  - Zero Critical/High defects open

8. RISK MITIGATION PLAN

Risk	Impact	Mitigation Strategy	Owner
Complex pricing logic errors	Financial loss, client disputes	Early prototyping of pricing rules; pe	CPQ Config Lead
SAP reconciliation failures	Financial reporting errors	Daily reconciliation scripts in SIT; mo	Integration Architect
DocuSign sync failures	Contract execution delays	Implement retry logic + alerting; test	DevOps Engineer
GDPR non-compliance	Regulatory fines	Automated PII detection scans; man	Security Officer
Quarter-end performance degradati	Business disruption	Performance baseline pre-UAT; load	Performance Engineer

9. DELIVERABLES & TIMELINE

9.1 Key Test Deliverables

- Test Strategy Document (this document) ✅
- Detailed Test Plans per module (Configuration, Pricing, Integration)
- Automated Test Scripts Repository (Provar)
- Test Data Catalog (masked + synthetic datasets)
- Performance Baseline Report
- UAT Execution Report & Sign-off Package
- Post-Implementation Validation Checklist

9.2 High-Level Timeline (Assuming 12-Week Enhancement Cycle)

Week 1-2: Test Planning + Environment Prep

Week 3-5: Component/Automation Development + SIT Execution

Week 6-7: Integration Testing + Performance Baseline

Week 8-9: UAT Execution + Compliance Validation

Week 10: Defect Retest + Sign-off Preparation

Week 11: Pre-Prod Smoke + Go/No-Go

Week 12: Production Deployment + Hypercare Support

10. ASSUMPTIONS & DEPENDENCIES

Assumptions

- Business SMEs available for UAT execution within scheduled windows
- SAP test environment available with mock posting capabilities for SIT
- DocuSign sandbox environment supports status webhook simulation
- GDPR masking rules finalized before SIT data refresh

Dependencies

- CPQ configuration freeze 1 week before SIT entry
- API contracts (REST/SOAP) documented and versioned prior to integration testing
- Jira workflow configured to auto-trigger Provar tests on deployment promotion

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🔍 SENIOR TEST ARCHITECT REVIEW & RECOMMENDATIONS

✅ Strengths of This Strategy

1. Risk-Aligned Prioritization: Focusing on quote calculation accuracy, integration integrity, and audit trails directly addresses the highest financial and compliance risks in wealth management.

2. Pragmatic Automation Scaling: Targeting 60% automation (from 30%) with Provar is realistic and leverages existing investment. The 60/30/10 pyramid balances maintenance cost with coverage.

3. Shift-Left Quality Gates: Auto-triggered tests on Jira deployment promotions enable early defect detection, reducing UAT churn.

4. GDPR-First TDM Approach: Combining Salesforce native masking with synthetic data generation satisfies compliance while enabling robust edge-case testing.

5. Clear Exit Criteria: "Zero Critical/High defects" with business sign-off creates unambiguous release readiness criteria.

⚠ Critical Recommendations for Execution Success

1. RBAC Validation Matrix Must Be Explicit

> While I recommended 8-10 roles, document the exact permission set combinations to test. Example:

> - Advisor: Can create quotes only for assigned clients

> - Senior Advisor: Can approve quotes <$1M

> - Compliance Officer: Can place holds but not modify pricing

> - Finance: Read-only access to finalized contracts for SAP reconciliation

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> Action: Create a RBAC test matrix in Week 1 with expected access outcomes per object/field.

2. Performance Testing Must Simulate Real Quarter-End Conditions

> A <10 minute SLA is generous, but wealth management quotes often involve complex calculations (FX rates, commission tiers, regulatory checks).

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> Action:

> - Instrument CPQ logs to identify calculation bottlenecks early

> - Test with 3x expected peak concurrency to build headroom

> - Include SAP mock response delays to validate timeout handling

3. DocuSign Sync Failure Handling Needs Chaos Testing

> Integration failures during contract signing can cause legal/compliance exposure.

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> Action:

> - Implement a "failure injection" test suite: simulate DocuSign API timeouts, invalid envelope IDs, webhook delivery failures

> - Validate that CPQ retains quote state and provides clear user guidance during sync issues

4. Reconciliation Testing Requires Automated Validation Scripts

> Manual SAP ↔ CPQ reconciliation is error-prone and unscalable.

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> Action:

> - Develop SQL/SOQL scripts to compare: Quote Amount (CPQ) vs. Posted Revenue (SAP)

> - Run reconciliation checks as automated post-deployment smoke tests

> - Alert on variance >0.01% for investigation

5. Audit Trail Validation Should Be Automated Where Possible

> Manual review of audit logs is inefficient and risks missing gaps.

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> Action:

> - Use Provar to verify that key field changes (Price, Discount, Approver) generate audit entries

> - Validate audit log immutability: attempts to modify logs should be blocked and alerted

🚀 Final Architectural Guidance

1. Adopt a "Compliance by Design" Testing Mindset: Embed GDPR and audit checks into every test case template—not as a separate phase.

2. Invest in Test Observability: Instrument tests to capture not just pass/fail, but performance metrics, API response times, and data lineage. This accelerates root-cause analysis.

3. Plan for Hypercare Testing: Post-go-live, have a lightweight regression suite ready to validate production fixes without full UAT cycles.

4. Document Lessons Learned: After UAT, capture which high-risk scenarios actually manifested defects. Refine the risk matrix for the next enhancement cycle.

> **Architect's Sign-off Recommendation:**

> This Test Strategy is APPROVED FOR EXECUTION contingent on:

> 1. Finalizing the RBAC test matrix by Week 1

> 2. Confirming SAP mock service availability for SIT

> 3. Validating Provar license capacity for 60% automation target

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> With these prerequisites met, this strategy provides a robust, compliant, and risk-focused foundation for delivering a high-quality Salesforce CPQ enhancement in the wealth management domain.

Senior Test Architect

Quality is not an act, it is a habit — especially when client wealth and regulatory compliance are on the line. 💼🔐✅